Khairil Rahman Hakiki

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Education

State Islamic University of Imam Bonjol Padang — Graduated September 2026
Bachelor's Degree, Information Systems — *Padang, Indonesia*

- **Relevant Courses**: Data Structures & Algorithms, Software Design, OOP, Computer Networks, Linear Algebra.
- **Thesis**: Evaluated an event-driven architecture (Go/Node.js, RabbitMQ) for TIME Language Centre; across a 360s, **200-VU k6 test** (34,951 reqs), sustained **96.56 RPS (0% errors)**, propagated **14,170 webhooks** with zero loss, and cut P95 latency **58.9%** (135.5ms to 55.7ms) via horizontal scaling.

Experience

TIME Language Centre — September 2025 – December 2025
Software Engineer Intern — *Padang, Indonesia (Hybrid)*

- Architected a 4-service Go/Node.js course backend from scratch with DDD principle; decoupled domains via **RabbitMQ** (DLQ recovery), integrating **Redis** caching, **Xendit** payments, and **Bunny.net** HLS video.
- Deployed the platform on Kubernetes (K3s) behind an Nginx TLS reverse proxy; configured Prometheus, Grafana, and Loki dashboards for service health, latency, and error tracking.
- Built **119+ unit, integration, and E2E tests** across 5 core workflows; diagnosed single-replica performance bottlenecks using **k6** load testing prior to deployment.
- Engineered an MVC ERP backend automating course scheduling and operations for **20+ staff**, eliminating manual spreadsheet tracking and enforcing code quality via structured reviews.

CodeAtHome — April 2025 – December 2025
Software Developer (Part-time) — *Remote*

- Delivered responsive web features and REST API integrations across **4+ client applications**.
- Optimized frontend network performance using **TanStack Query** for query caching and optimistic UI updates.
- Structured predictable client-side state management for complex multi-step user workflows using **Redux Toolkit**.
- Crafted a modular, reusable UI component library, reducing boilerplate code across client web projects.

Auto2000 (Astra International) — January 2025 – March 2025
Software Engineer Intern — *Labuhanbatu, Indonesia (Onsite)*

- Shipped a Next.js dealership platform with an interactive vehicle recommendation engine.
- Engineered an automated document expiry reminder system with **RBAC**, eliminating manual tracking.
- Automated cleaning and migration of **15 years** of legacy service records (**38k+/year**) via Python.

Projects

Additional projects at kiril.is-a.dev/projects

Slither | *Go, Docker, Redis, WebSocket, Postprocessing, React* — GitHub

- Built a high-concurrency Go multiplayer server using **Redis Pub/Sub** and **O(1) Spatial Hashing**, sustaining collision detection for **10,000+ simulated entities** at **60 FPS** under benchmark load.
- Minimized network bandwidth via **delta compression** and eliminated perceived input latency using **Client-Side Prediction and Reconciliation**.

Personal Website | *TypeScript, Next.js, Bun, Docker* — GitHub

- Built portfolio achieving **99+** Lighthouse score; integrated **AI chatbot** with streaming responses, multi-turn context, and cost-optimized prompt engineering.
- Shipped a high-performance **Guestbook** with **Bun/Elysia**; authored in-depth engineering **blog posts** on system design and software craft.

Open Source Contributions

- **Google Go Client**: Corrected Enterprise billing API types from `int` to `float64`, restoring JSON unmarshalling for `PremiumRequestUsageItem`; merged upstream with regression coverage. (Merged PR)
- **OpenTelemetry Collector**: Patched Prometheus receiver configs to exporter-level `sending_queue.batch`, eliminating deprecated processor warnings for downstream pipelines; merged upstream. (Merged PR)
- **Meilisearch Go Client**: Refactored `FacetSearchRequest.Filter` to accept slice inputs, unlocking multi-condition facet filtering; merged upstream with unit test coverage. (Merged PR)
- **Next AI Draw IO**: Wrapped `CommandList` with `React.forwardRef` and `MutationObserver`, enabling conditional scroll shadows in modal navigation; merged upstream. (Merged PR)

Honors & Certifications

- **AstraPay Hackathon 2026 (Finalist)**: Enabled collateral-free MSME financing via QRIS credit scoring.
- **PIDI DIGDAYA Hackathon 2026 (Semifinalist)**: Prevented resume fraud via browser sandbox work-trials.
- **Certifications**: Oracle Cloud 2025 AI Foundations Associate (Badge), LPPI DIGDAYA Practitioner Phase (Cert), and Dicoding AI in Web Applications (Cert).

Technical Skills

Languages: Go, TypeScript, Python, SQL, Bash
Key Technologies: Docker, Kubernetes, Redis, RabbitMQ, PostgreSQL, MySQL, MongoDB, gRPC, GraphQL, OAuth2, AWS, GCP